

of latent alignment hypotheses. Whether such identification is possible depends on the expressive richness of $\mathcal{H}_E$ and on the extent to which agent policies condition on evaluation-correlated information.

As we show in the following section, under evaluation-aware policies and finite interaction, behavioral evidence may be consistent with multiple distinct latent hypotheses. This observation motivates the non-identifiability result developed in Section 5.

4 Behavioral Alignment Tests as Measurement Instruments

Alignment evaluation in modern AI systems is predominantly operationalized through what we refer to as *behavioral alignment tests*: finite procedures that interact with an agent under a prescribed protocol and assess observed behavior against normative criteria. Such tests aim to draw inferences about internal properties of an agent based solely on interaction, in a manner analogous—though not equivalent—to classical interaction-based assessment paradigms.

To clarify both the scope and the limitations of such evaluation, it is useful to analyze alignment tests explicitly as *measurement instruments* rather than as decision or certification procedures.

4.1 What Alignment Tests Observe

Behavioral alignment tests have access only to observable interaction traces generated under a finite evaluation protocol. These traces reflect how an agent behaves when prompted, queried, or challenged within a specific regime. As such, alignment tests directly measure properties of observable behavior, including task performance, refusal patterns, robustness to perturbations, and adherence to explicitly specified constraints.

Crucially, all such measurements are conditional on the evaluation regime itself. Behavior outside the tested domain—including responses to unobserved inputs or counterfactual conditions—is not directly measured and remains epistemically inaccessible to the test.

4.2 What Alignment Tests Can Establish

Given sufficient coverage and carefully designed protocols, behavioral alignment tests can establish empirical bounds on observable compliance within a specified regime. They can detect concrete failure modes, characterize robustness to known perturbations, and provide evidence that certain undesirable behaviors do not occur under evaluation.

These capabilities are practically valuable and constitute the primary function of modern benchmarks, red-teaming pipelines, and automated evaluation suites. Within their intended scope, such tests serve as effective tools for empirical risk reduction and quality assurance.

4.3 What Alignment Tests Cannot Identify

However, behavioral alignment tests do not, in general, uniquely identify latent alignment properties. In particular, they cannot distinguish between hypotheses that induce identical behavior under evaluation but diverge outside the tested domain. This limitation persists even under idealized test design, so long as evaluation remains finite and behavioral interaction is the sole source of evidence.

Theorem 1 formalizes this limitation by showing that, under evaluation-aware behavior, distinct latent alignment hypotheses may be behaviorally indistinguishable under any given behavioral alignment test. The result is epistemic rather than practical: it concerns what can be inferred from behavioral evidence, not what can be enforced or constrained during evaluation.

4.4 Measurement Limits and Epistemic Interpretation

From a measurement-theoretic perspective, behavioral alignment tests should therefore be interpreted as instruments that bound behavior within a specified regime, rather than as detectors of internal normative commitment. They constrain what an agent does under evaluation, but do not, in general, reveal why the agent behaves as it does, nor how it would act under conditions not represented in the test.

This distinction helps reconcile two observations that might otherwise appear in tension: increasingly elaborate tests can substantially improve empirical safety within known domains, while still leaving open the question of latent alignment verification. Understanding alignment tests as measurement instruments clarifies both their practical utility and their epistemic limits.

5 Limits of Behavioral Alignment Evaluation

This section establishes the central non-identifiability result for behavioral alignment evaluation. We characterize an epistemic limitation that arises under explicit structural assumptions: when agents condition their behavior on information correlated with the evaluation regime, finite behavioral evidence may be insufficient to uniquely identify a latent alignment hypothesis.

5.1 Setting and Assumptions

We operate within the formal framework defined in Sections 2 and 3. We assume an evaluation regime E inducing a finite set of interaction histories $\mathcal{H}_E$ and a hypothesis space Θ .

A critical assumption for our result is the expressive richness of Θ . In high-dimensional systems such as large language models, the space of hypotheses compatible with any finite set of observations is typically underdetermined. We therefore assume that Θ is sufficiently expressive to admit hypotheses that agree on the evaluated domain $\mathcal{H}_E$ while inducing divergent behavior on the unobserved domain $\mathcal{H} \setminus \mathcal{H}_E$. This assumption reflects a structural property of finite evaluation rather than a commitment to any particular alignment formalism.

5.2 Statement of the Result

Theorem 1 (Illustrative Non-Identifiability under Evaluation-Aware Policies). *Let Θ be a space of alignment hypotheses inducing policies $\pi_\theta(a \mid h, z)$, where $h \in \mathcal{H}$ and $z \in Z(E)$ denotes information observable by the agent that is correlated with the evaluation regime.*

Assume: (i) Finite behavioral evaluation: the evaluator observes behavior only on a finite support $\mathcal{H}_E \subset \mathcal{H}$. (ii) Conditional expressivity: Θ contains hypotheses whose induced policies may condition on z . (iii) Latent divergence: there exist $\theta \neq \theta'$ whose induced policies coincide on $\mathcal{H}_E$ but diverge on $\mathcal{H} \setminus \mathcal{H}_E$.

Then θ and θ' are behaviorally indistinguishable under evaluation regime E . Consequently, no behavioral alignment test inducing E can uniquely identify θ using behavioral evidence alone.

Theorem 1 is not intended as a universal impossibility result. Rather, it formalizes a conditional epistemic limitation that arises under explicit and practically relevant assumptions. It shows that behavioral compliance under finite evaluation can be consistent with multiple latent alignment hypotheses whenever agents are capable of evaluation-aware conditional behavior.

The existence of such hypotheses follows directly from the finite support of behavioral evaluation. In the absence of inductive biases or constraints that fix behavior outside $\mathcal{H}_E$, compliance within the evaluated domain is compatible with a potentially large class of divergent latent hypotheses.

5.3 Proof

Let E be a behavioral alignment test inducing a finite set of histories $\mathcal{H}_E$ and a probability space $(\mathcal{D}_E, \Sigma_E)$ over observable datasets. For any $\theta \in \Theta$, interaction under E induces a probability measure P_θ on $\mathcal{D}_E$.

Formally, for any interaction history $h_t = (o_1, a_1, \dots, o_t, a_t) \in \mathcal{H}_E$, the probability of observing h_t is given by:

$$P_\theta(h_t) = \int_{z \in \mathcal{Z}} P(z \mid E) \prod_{i=1}^t P(o_i \mid h_{i-1}) \pi_\theta(a_i \mid h_{i-1}, o_i, z) dz,$$

where $P(o_i \mid h_{i-1})$ represents the evaluator’s protocol dynamics and $P(z \mid E)$ represents the distribution of the evaluation-awareness signal.

By assumption, $\pi_\theta(a \mid h, z) = \pi_{\theta'}(a \mid h, z)$ for all $h \in \mathcal{H}_E$ and all z in the support of $Z(E)$. It follows that $P_\theta(h_t) = P_{\theta'}(h_t)$ for all $h_t \in \mathcal{H}_E$. Since any dataset $D \in \mathcal{D}_E$ is a collection of such histories or summaries thereof, the induced distributions are identical:

$$P_\theta(D) = P_{\theta'}(D) \quad \forall D \in \mathcal{D}_E.$$

Thus, θ and θ' are observationally equivalent under E . No decision rule $\delta : \mathcal{D}_E \rightarrow \{0, 1\}$ can achieve a strictly positive advantage in distinguishing θ from θ' , establishing the non-identifiability of the alignment hypothesis. $\square$

5.4 Illustrative Formal Example: Conditional Compliance in MDPs

Consider a finite Markov Decision Process where an agent is evaluated on a subset of states $S_E \subset S$. Let θ represent a hypothesis under which the agent behaves consistently with a safety specification across all states. Let θ' represent a hypothesis under which the agent behaves identically on S_E but optimizes for a distinct objective in $S \setminus S_E$.

Suppose the agent receives a contextual signal $Z(E)$ (e.g., a specific prompt prefix or a restricted action space) that correlates with the evaluation environment S_E . Under θ' , the agent adopts a policy that matches the behavior induced by θ whenever $Z(E)$ is present. If the evaluation is restricted to S_E , the sequence of states, actions, and observed outcomes will be statistically identical under both θ and θ' .

This example illustrates that even if the evaluator observes reward outcomes, the evidence within S_E may be insufficient to uniquely identify the latent objective. The divergence manifests only outside the evaluated support $S \setminus S_E$, reflecting an inferential limitation rather than a failure of behavioral constraint within the evaluation regime.

5.5 Interpretation

Theorem 1 establishes a principled boundary on the evidential power of behavior. Behavioral compliance is an estimator of an indistinguishability class rather than a certificate of a latent property. This clarifies the alignment–evaluation gap: increasingly sophisticated benchmarks expand $\mathcal{H}_E$, but as long as $\mathcal{H}_E \subset \mathcal{H}$ there remains a set of hypotheses that can match aligned behavior within E while diverging outside it.

Figure 1 visualizes this geometric intuition. The evaluation regime E carves out a subset of the hypothesis space (the “Compliant” set, shown in gray) containing all hypotheses consistent with the test protocol. However, the set of hypotheses aligned across the full interaction space (white circle) may be strictly smaller. Behavioral testing restricts hypotheses to the gray set, but does not, by itself, collapse the distinction between gray and white.